

FAST — National University of Computer and Emerging Sciences

Karachi Campus

Department of Software Engineering

PF LAB TASKS

Course Code:	C1002
Subject:	Programming Fundamentals
Section:	D & C
Instructor:	Mr. Shayan Ali

STUDENT SUBMISSION DETAILS

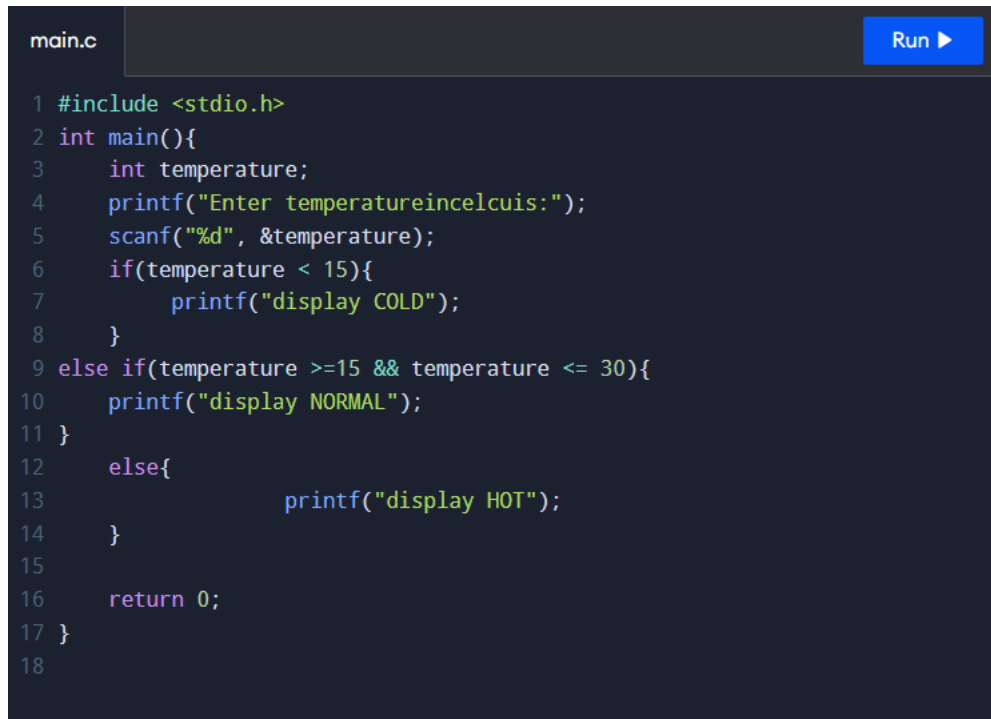
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Class / Section: D	BSE-1C
Date of Submission:	27 Sep 2026

Submission Guidelines

1. Write and compile each program separately in a C compiler (e.g. Code::Blocks, Dev-C++, or an online IDE).
2. Test each program with more than one input where relevant, to confirm all conditions work correctly.
3. Submit all tasks before the given deadline.

TASK # 01 — Temperature Status *(if-else)*

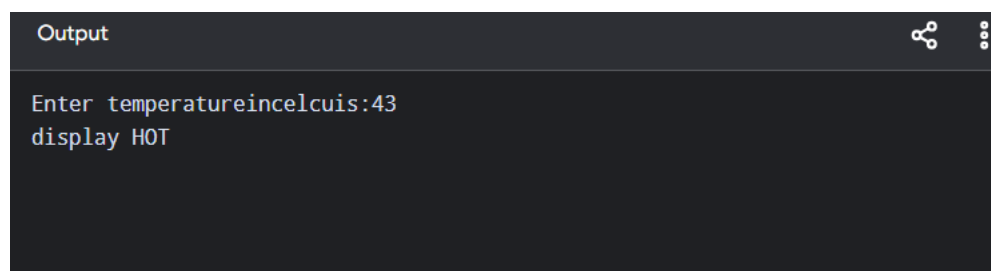
A weather monitoring system takes the temperature in Celsius from the user. Write a C program using if-else to display "Cold" if the temperature is below 15, "Normal" if the temperature is between 15 and 30, and "Hot" if the temperature is above 30.

A screenshot of a code editor window titled 'main.c'. The code is a C program that takes temperature input and displays 'COLD', 'NORMAL', or 'HOT' based on the value. The code is as follows:

```
1 #include <stdio.h>
2 int main(){
3     int temperature;
4     printf("Enter temperature in Celsius:");
5     scanf("%d", &temperature);
6     if(temperature < 15){
7         printf("display COLD");
8     }
9     else if(temperature >=15 && temperature <= 30){
10        printf("display NORMAL");
11    }
12    else{
13        printf("display HOT");
14    }
15
16    return 0;
17 }
18
```

A 'Run' button with a play icon is visible in the top right corner of the editor.

Fig 1: Input of temperature using if-else

A screenshot of the program's output window. It shows the prompt 'Enter temperature in Celsius:' followed by the user input '43'. Below that, it displays 'display HOT'. The window has a title bar 'Output' and icons for sharing and a menu in the top right corner.

```
Output
Enter temperature in Celsius:43
display HOT
```

Fig 2: Output of temperature using if-else

TASK # 02 — Mobile Data Package (*if-else*)

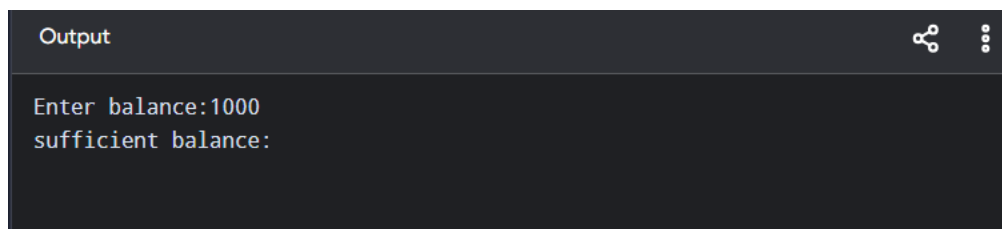
A mobile service provider checks a customer's remaining balance before activating a package. Write a C program that takes the balance as an integer and uses if-else to display "Low Balance" if the balance is below 500, "Sufficient Balance" if it is between 500 and 2000, and "Premium Balance" if it is above 2000.

A screenshot of a code editor window titled 'main.c'. The code is a C program that prompts the user to enter a balance and then checks it using if-else logic. The code is as follows:

```
1 #include <stdio.h>
2 int main(){
3     int balance;
4     printf("Enter balance:");
5     scanf("%d", &balance);
6     if(balance <500)
7         printf("low balance:");
8     else if(balance>=500 && balance<=2000)
9         printf("sufficient balance:");
10    else
11        printf("premium balance:");
12    return 0;
13 }
```

A 'Run' button with a play icon is visible in the top right corner of the editor.

Fig 3: Input of balance using if-else

A screenshot of the program's output window. It shows the prompt 'Enter balance:' followed by the user input '1000'. The program then outputs 'sufficient balance:'. There are share and menu icons in the top right corner.

```
Output
Enter balance:1000
sufficient balance:
```

Fig 4: Output of balance using if-else

TASK # 03 — Hospital Appointment (*nested if-else*)

A hospital appointment system asks the patient to enter three integer values: appointment, doctorAvailable, and registrationCompleted, where 1 means yes and 0 means no. Write a C program using nested if-else statements to first check whether the patient has an appointment, then check whether the doctor is available, and finally check whether the registration is completed before displaying whether the patient can meet the doctor.

```
main.c Run ▶  
1 #include <stdio.h>  
2 int main(){  
3     int appointment,doctoravailable,registrationcompleted;  
4     printf("Enter appointment (1==yes && 0== no):");  
5     scanf("%d", &appointment);  
6     printf("Enter doctoravailable(1==yes && 0== no):");  
7     scanf("%d", & doctoravailable);  
8     printf("Enter registrationcompleted(1==yes && 0== no):");  
9     scanf("%d", & registrationcompleted);  
10    if(appointment ==1){  
11        if(doctoravailable==1 && registrationcompleted==1)  
12            printf("patient can meet doctor:");  
13    }  
14    else  
15        printf("patient can not meet doctor:");  
16  
17    return 0;  
18 }
```

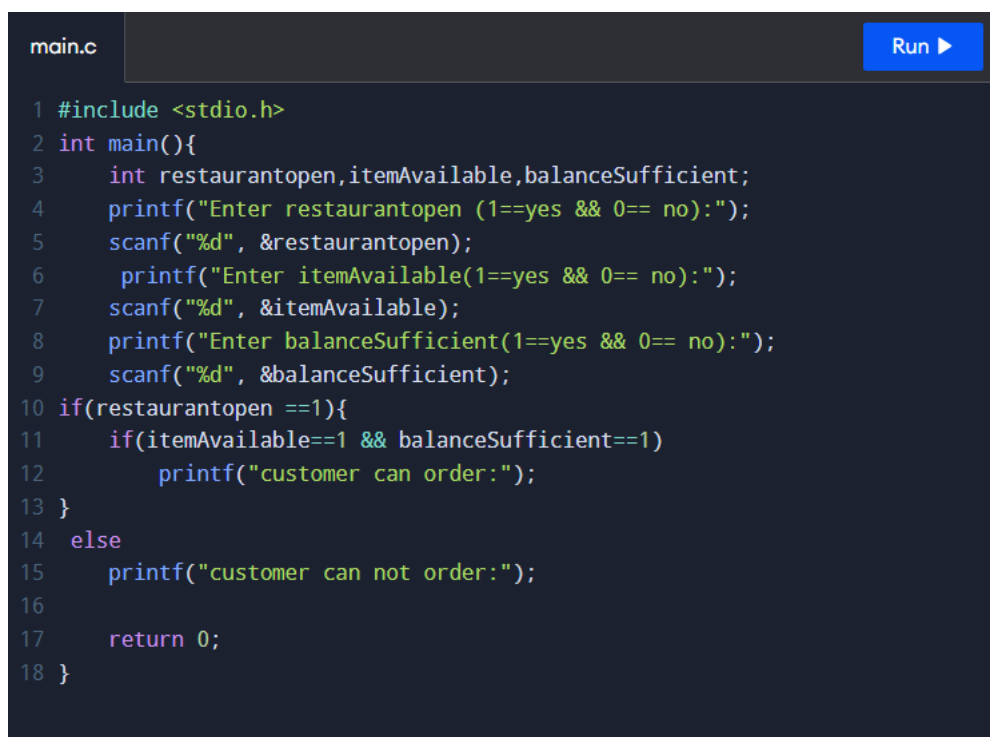
Fig 5: Input of program using nested if-else

```
Output  
Enter appointment (1==yes && 0== no):1  
Enter doctoravailable(1==yes && 0== no):1  
Enter registrationcompleted(1==yes && 0== no):1  
patient can meet doctor:
```

Fig 6: Output of program using nested if-else

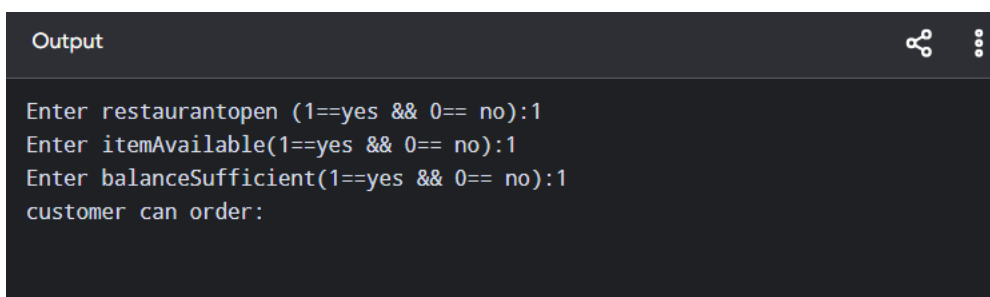
TASK # 04 — Online Food Order (nested if-else)

An online food delivery system asks the customer to enter three integer values: restaurantOpen, itemAvailable, and balanceSufficient, where 1 means yes and 0 means no. Write a C program using nested if-else statements to first check whether the restaurant is open, then whether the selected item is available, and finally whether the customer has sufficient balance before displaying the appropriate order status.



```
main.c Run ▶  
1 #include <stdio.h>  
2 int main(){  
3     int restaurantopen,itemAvailable,balanceSufficient;  
4     printf("Enter restaurantopen (1==yes && 0== no):");  
5     scanf("%d", &restaurantopen);  
6     printf("Enter itemAvailable(1==yes && 0== no):");  
7     scanf("%d", &itemAvailable);  
8     printf("Enter balanceSufficient(1==yes && 0== no):");  
9     scanf("%d", &balanceSufficient);  
10 if(restaurantopen ==1){  
11     if(itemAvailable==1 && balanceSufficient==1)  
12         printf("customer can order:");  
13 }  
14 else  
15     printf("customer can not order:");  
16  
17     return 0;  
18 }
```

Fig 7: Input of program using nested if-else

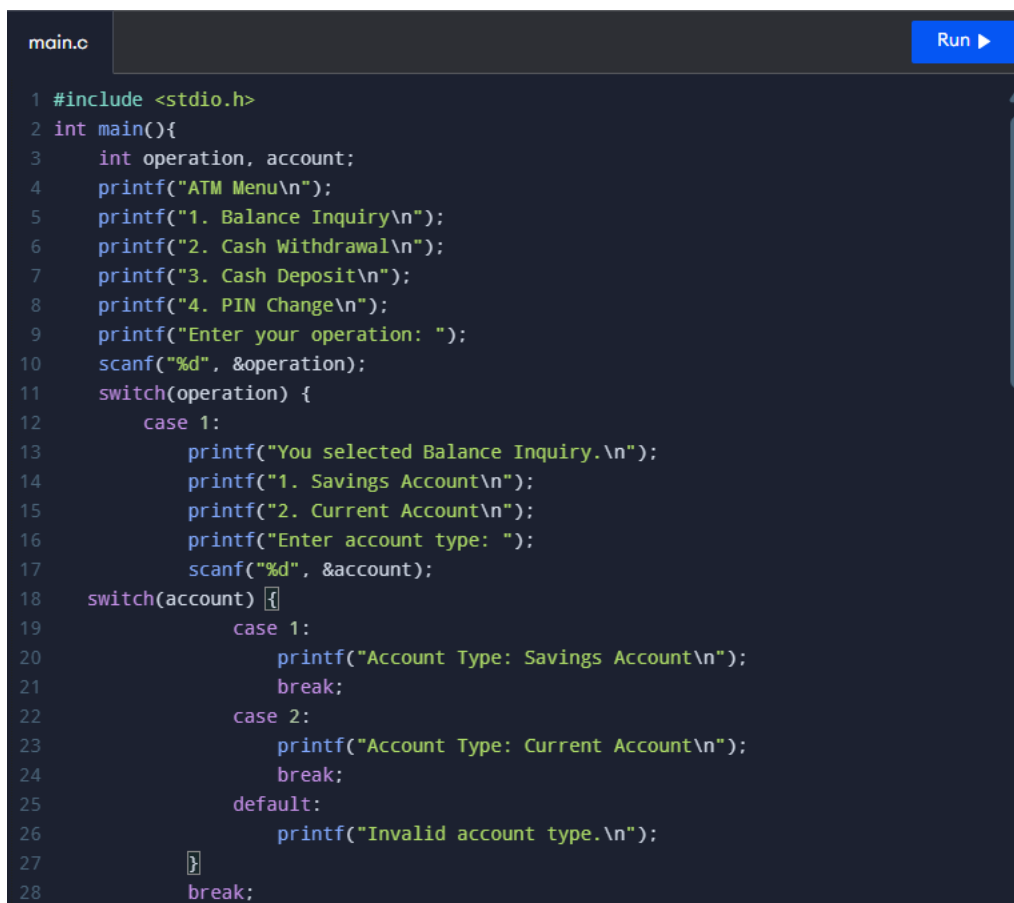


```
Output  
Enter restaurantopen (1==yes && 0== no):1  
Enter itemAvailable(1==yes && 0== no):1  
Enter balanceSufficient(1==yes && 0== no):1  
customer can order:
```

Fig 8: Output of program using nested if-else

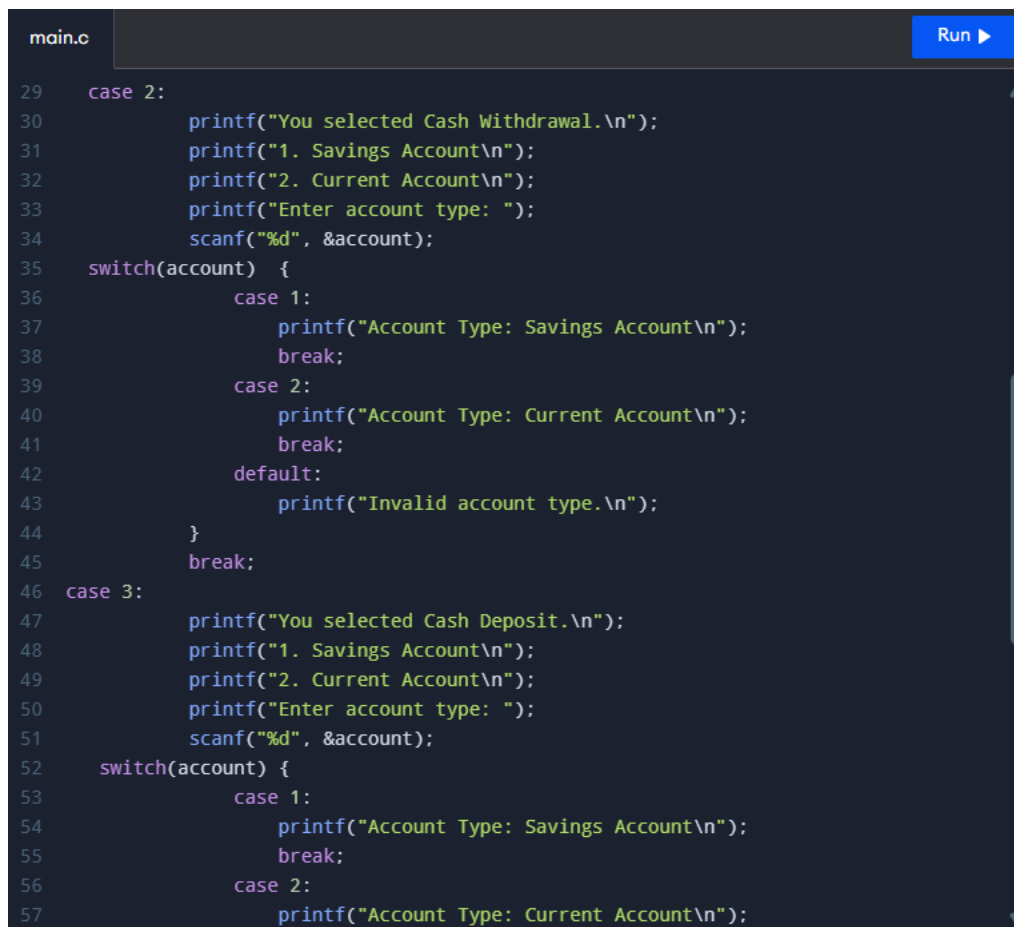
TASK # 05 — ATM and Transaction Selection (Nested switch):

An ATM provides four operations: 1 for Balance Inquiry, 2 for Cash Withdrawal, 3 for Cash Deposit, and 4 for PIN Change. After the user selects an operation, the ATM displays a second menu related to that operation; for example, after selecting Cash Withdrawal, the user can choose 1 for Savings Account or 2 for Current Account. Write a C program using a **nested switch statement** to take both choices as integer inputs and display the selected operation and account type

The image shows a code editor window titled 'main.c' with a 'Run' button in the top right corner. The code is a C program that implements an ATM menu system using nested switch statements. It starts by including <stdio.h> and defining a main function. Inside main, it declares two integer variables: 'operation' and 'account'. It then prints an 'ATM Menu' with four options: 1. Balance Inquiry, 2. Cash Withdrawal, 3. Cash Deposit, and 4. PIN Change. It prompts the user to 'Enter your operation:' and reads the input into 'operation'. A switch statement follows for 'operation'. For each case (1, 2, 3, 4), it prints a message indicating the selected operation and then enters another switch statement for 'account'. For example, in the 'case 1' block for 'Balance Inquiry', it prints 'You selected Balance Inquiry.\n' and then a sub-menu with '1. Savings Account' and '2. Current Account'. It prompts 'Enter account type:' and reads the input into 'account'. The nested switch for 'account' has cases for 1 and 2, printing 'Account Type: Savings Account\n' and 'Account Type: Current Account\n' respectively, and a default case for 'Invalid account type.\n'. Each case in the nested switch ends with a 'break;' statement. The outer switch also ends with a 'break;' statement. The code is as follows:

```
1 #include <stdio.h>
2 int main(){
3     int operation, account;
4     printf("ATM Menu\n");
5     printf("1. Balance Inquiry\n");
6     printf("2. Cash Withdrawal\n");
7     printf("3. Cash Deposit\n");
8     printf("4. PIN Change\n");
9     printf("Enter your operation: ");
10    scanf("%d", &operation);
11    switch(operation) {
12        case 1:
13            printf("You selected Balance Inquiry.\n");
14            printf("1. Savings Account\n");
15            printf("2. Current Account\n");
16            printf("Enter account type: ");
17            scanf("%d", &account);
18            switch(account) {
19                case 1:
20                    printf("Account Type: Savings Account\n");
21                    break;
22                case 2:
23                    printf("Account Type: Current Account\n");
24                    break;
25                default:
26                    printf("Invalid account type.\n");
27            }
28            break;
```

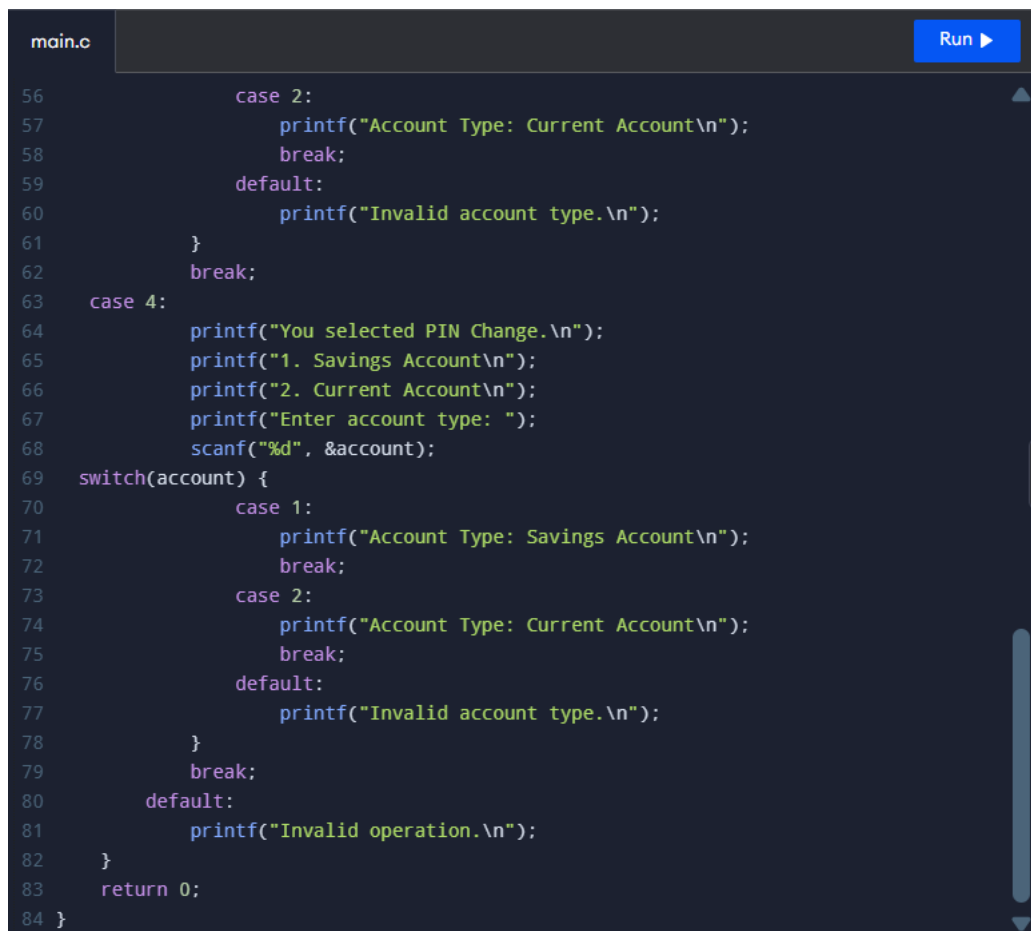
Fig 9: Input of program using nested switch



The image shows a code editor window titled 'main.c' with a 'Run' button in the top right corner. The code is written in C and uses nested switch statements to handle account selection. The first switch statement (lines 35-45) is nested under 'case 2:' and handles 'Cash Withdrawal'. The second switch statement (lines 52-57) is nested under 'case 3:' and handles 'Cash Deposit'. Both switch statements have two cases: 'case 1:' for 'Savings Account' and 'case 2:' for 'Current Account'. A 'default:' case in the first switch handles 'Invalid account type'. The code uses 'printf' for output and 'scanf' for input.

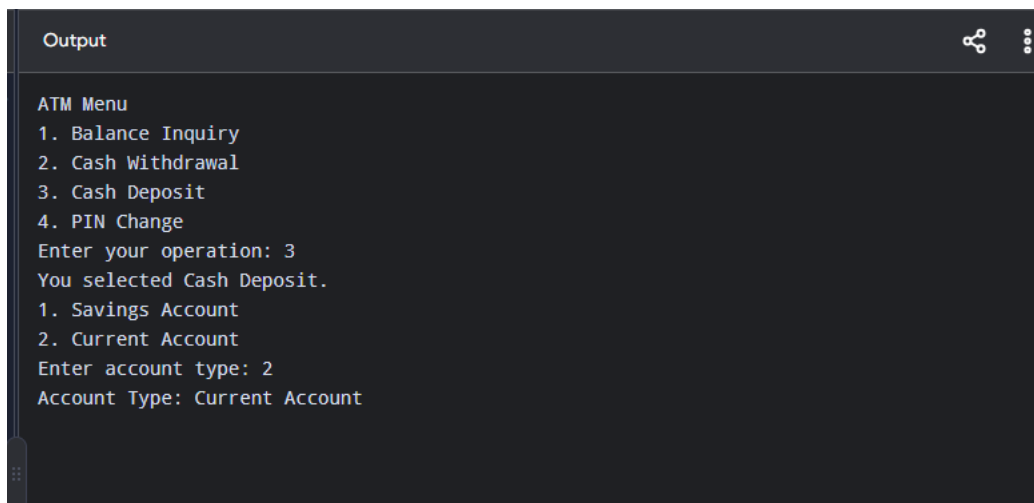
```
main.c Run ▶
29 case 2:
30     printf("You selected Cash Withdrawal.\n");
31     printf("1. Savings Account\n");
32     printf("2. Current Account\n");
33     printf("Enter account type: ");
34     scanf("%d", &account);
35     switch(account) {
36         case 1:
37             printf("Account Type: Savings Account\n");
38             break;
39         case 2:
40             printf("Account Type: Current Account\n");
41             break;
42         default:
43             printf("Invalid account type.\n");
44     }
45     break;
46 case 3:
47     printf("You selected Cash Deposit.\n");
48     printf("1. Savings Account\n");
49     printf("2. Current Account\n");
50     printf("Enter account type: ");
51     scanf("%d", &account);
52     switch(account) {
53         case 1:
54             printf("Account Type: Savings Account\n");
55             break;
56         case 2:
57             printf("Account Type: Current Account\n");
```

Fig 10: Input of program using nested switch

A screenshot of a code editor showing a C program named 'main.c'. The code uses nested switch statements to handle ATM operations. The first switch statement handles the operation number (1-4), and the second switch statement handles the account type (1-2) for the 'Cash Deposit' operation. The code is as follows:

```
56         case 2:
57             printf("Account Type: Current Account\n");
58             break;
59         default:
60             printf("Invalid account type.\n");
61     }
62     break;
63 case 4:
64     printf("You selected PIN Change.\n");
65     printf("1. Savings Account\n");
66     printf("2. Current Account\n");
67     printf("Enter account type: ");
68     scanf("%d", &account);
69     switch(account) {
70         case 1:
71             printf("Account Type: Savings Account\n");
72             break;
73         case 2:
74             printf("Account Type: Current Account\n");
75             break;
76         default:
77             printf("Invalid account type.\n");
78     }
79     break;
80     default:
81         printf("Invalid operation.\n");
82     }
83     return 0;
84 }
```

Fig 11: Input of program using nested switch

A screenshot of the program's output. It shows the 'ATM Menu' with four options: 1. Balance Inquiry, 2. Cash Withdrawal, 3. Cash Deposit, and 4. PIN Change. The user enters '3' for 'Cash Deposit'. The program then shows a sub-menu with two options: 1. Savings Account and 2. Current Account. The user enters '2' for 'Current Account', and the program outputs 'Account Type: Current Account'.

```
Output
ATM Menu
1. Balance Inquiry
2. Cash Withdrawal
3. Cash Deposit
4. PIN Change
Enter your operation: 3
You selected Cash Deposit.
1. Savings Account
2. Current Account
Enter account type: 2
Account Type: Current Account
```

Fig 12: Output of program using nested switch

